DEPARTMENT OF MINERALS AND ENERGY
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DIVISION OF NATIONAL MAPPING
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THE ADJUSTMENT OF THE AUSTRALIAN LEVELLING SURVEY 1970-1971

by

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4.4.7 The special calibrating device was devised and built at the Flinders University of South Australia. It consisted of a clockwork driven rotating chart drum with two recording pens. Each pen was connected by insulated leads to their activating pressure sensor heads. The sensor heads detected changes in the pressure through sensitive membranes and could determine the height of water above each sensor. One sensor was placed near the bottom inside the tide gauge float well and the other outside the well at the same height as the inner sensor. From the resulting chart the differences in height between the water inside and outside the well could be compared and frequency responses determined.

4.4.8 Results of the 1967 tide gauge survey are contained in Survey Paper 6, "A Handbook of Selected Australian Tide Gauges" by A. K. Easton of the Horace Lamb Centre for Oceanographical Research and in Research Paper 24, "The Frequency Response of Selected Australian Tide Gauges" by A. K. Easton.

4.4.9 One problem found common to a large number of the remoter gauges visited was vandalism. Bullet holes were plentiful. Vandalism reached a climax when, in 1968, the gauge at Eucla was stolen complete with 3 months of data.

4.4.10 During 1968 and 1969 the Division of National Mapping completed the connection of a number of gauges to the levelling network where this connection from the three tide gauge bench marks to a levelling traverse had been too long or too difficult for the tide gauge survey party to undertake.

4.5 Analysis and Evaluation of Tide Gauge Records

4.5.1 Copies of the recorder charts from all tide gauges in the programme were sent by the operators to the Horace Lamb Centre for Oceanographical Research at the Flinders University of South Australia where hourly heights of sea level were extracted and all computations of mean sea levels and tidal constants were carried out.

4.5.2 A complete report of the results of the 1966-1968 tide gauge programme is given in "Tidal Program 1966-1968, Memorandum 5" by Alan Easton and Rainer Radok of the Horace Lamb Centre.

4.5.3 Mean sea level at Karumba was determined by the Queensland Department of Harbours and Marine from recordings made by them during the four year period 1957-1960.

4.5.4 Details of tide gauge bench marks, mean sea level values adopted for the determination of the Australian Height Datum and the height of the tide gauge bench marks above mean sea level at each station are given in Table 3 on page 36.

TABLE 3

HEIGHT OF MEAN SEA LEVEL ABOVE ZERO OF STAFF GAUGE, TIDE GAUGE BENCH MARKS AND HEIGHT OF BENCH MARKS ABOVE MEAN SEA LEVEL

Station	Tide Gauge Bench Mark	Height of Mean Sea Level above Zero of Staff Gauge (Feet)	Height of Bench Mark above Mean Sea Level (Feet
Port Macdonnell	NMV/E/11	2.010	6.908
Victor Harbour	NMV/E/13	1.900	12.645
Port Lincoln	NMV/E/16	3.330	11.718
Thevenard	NMV/E/18	3.250	15.863
Eucla	PWD BM	2.370	13.305
Esperance	NMV/F/2	1.820	11.521
Albany	HR 2	2.530	10.640
Bunbury	NMV/F/5	2.100	7.231
Fremantle	NMV/F/6	2.350	10.659
Geraldton	NMV/F/7	2.780	6.041
Carnarvon	NMV/F/9	2.670	11.115
Port Hedland	PWD PH1	10.350	14.889
Broome	EP 1	14.640	29.055
Wyndham	NMV/F/11	13.760	19.605
Darwin	BM 252	13.050	20.985
Centre Island	AHS 1	9.820	4.920
Karumba	SBM 10222	5.360	15.170
Weipa	NMV/B/304A	5.080	4.510
Bamaga	PBM 3931	4.500	10.160
Cooktown	NMV/B/307	4.370	22.912
Cairns	PSM 16253	4.700	10.910
Townsville	NMV/B/309	5.350	14.840
Mackay	NMV/B/311	9.650	25.110
Bundaberg	PSM 3853	4.420	14.335
Brisbane	TG BASE	3.590	18.610
Coffs Harbour	NAIL	2.680	18.350
Camp Cove	PM 84	3.040	4.983
Port Kembla	BM 6	2.880	13.710
Point Lonsdale	NW CNR	2.830	42.388
Port Fairy	TG BM	2.280	10.832

Epoch of mean sea level at Karumba 1.1.57 to 31.12.60.

Epoch for all other gauges 1.1.66 to 31.12.68.

6.6.2 The supplementary levelling was adjusted by LINADJ in the same way as the primary levelling (see 6.5) with two cards heading each section, punched with the junction point information. In this case the height of the JP was either an original LEVEL 2 adjusted height or the adjusted height of a bench mark on a primary traverse which the supplementary section was connected to.

6.6.3 The printed computer output of the supplementary levelling was dealt with in the same way as the primary levelling, both in the correction and distribution.

6.7 Metropolitan adjustment

6.7.1 So that the heights of bench marks within the metropolitan areas of Sydney and Perth need not change from their long established heights which are only trivially different from the newly adjusted values it was agreed at the 1971 National Mapping Council meeting that the old heights could be retained within clearly defined "metropolitan zones". These heights would be accepted as being on the AHD.

6.7.2 A "metropolitan zone" is the area, nominated by the State, in which the State desired to retain heights on a previously accepted datum. States were required to assign heights to all marks in this zone which were then quoted as being on the AHD.

6.7.3 The heights adopted in Sydney and Perth, although not directly computed as part of the National Levelling Adjustment, are sufficiently close for all practical purposes. The difference in heights determined from the adjustment and those adopted by the States in their metropolitan zones had to be absorbed in "buffer zones" in order to prevent abrupt datum changes at the edge of the metropolitan zones.

6.7.4 A "buffer zone" is the area, extending from the edge of the metropolitan zone generally to the next primary junction point, in which differences between State heights of "terminal bench marks" and NLA heights assigned to primary bench marks was distributed. The greater the distance between the metropolitan zone and the outer edge of the buffer zone the smaller the corrections between individual bench marks in the buffer zone. The adjustment of the levelling within the buffer zone will be undertaken by the Division of National Mapping.

6.7.5 "Terminal Bench Marks" are those marks at the edge of the metropolitan zone where level traverses passed from the metropolitan zone into the buffer zone. Heights of these marks were assigned by the States.